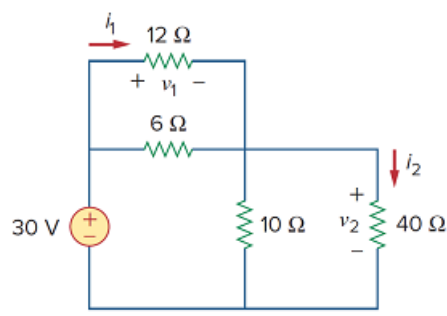


Problem

Find v_1 and v_2 in the circuit shown in Fig. 2.43. Also calculate i_1 and i_2 and the power dissipated in the $12\text{-}\Omega$ and $40\text{-}\Omega$ resistors.

Figure 2.43:



Step-by-step solution

Step 1 of 8

The circuit is shown in Figure 1.

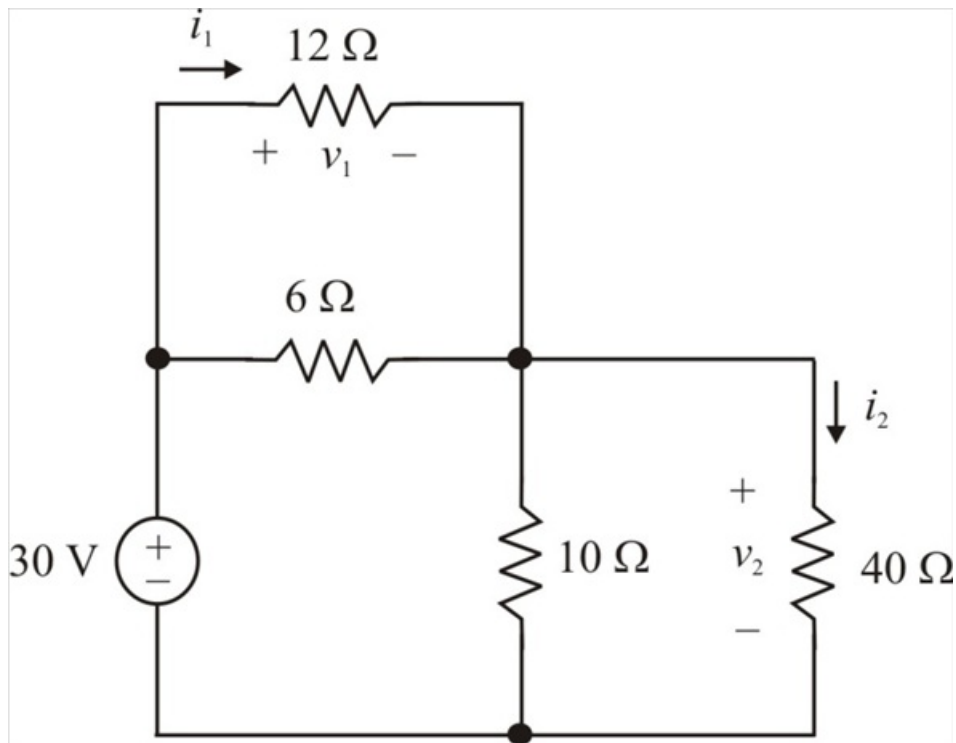


Figure 1

Step 2 of 8

The 12 Ω and 6 Ω resistors are in parallel. Their combined resistance is:

$$12\Omega \parallel 6\Omega = \frac{(12)(6)}{12+6} \\ = 4\Omega$$

The 10 Ω and 40 Ω resistors are in parallel. Their combined resistance is:

$$10\Omega \parallel 40\Omega = \frac{(10)(40)}{10+40} \\ = 8\Omega$$

Step 3 of 8

The circuit in Figure 1 is reduced to that in Figure 2.

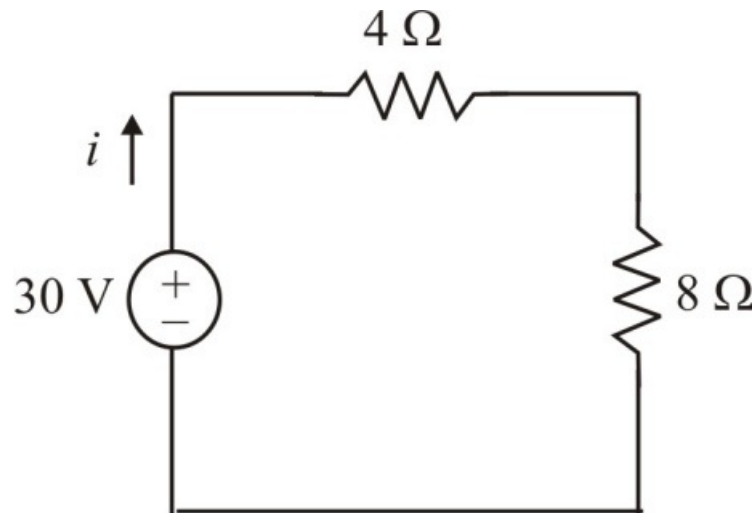


Figure 2

Step 4 of 8

Calculate the total current.

$$i = \frac{v}{R} \\ = \frac{30}{12} \\ = 2.5 \text{ A}$$

Apply current division.

$$i_1 = \frac{6}{6+12} i \\ = \frac{6}{6+12} (2.5) \\ = 833.3 \text{ mA}$$

Therefore, the current i_1 is 833.3 mA.

Step 5 of 8

By using Ohm's law,

$$\begin{aligned}v_1 &= i_1 R \\&= (833.3 \text{ mA})(12 \Omega) \\&= 10 \text{ V}\end{aligned}$$

Therefore, the voltage v_1 is 10 V.

Step 6 of 8

Calculate the power dissipated in 12 Ω resistor.

$$\begin{aligned}P_1 &= v_1 i_1 \\&= (10 \text{ V})(833.3 \text{ mA}) \\&= 8.333 \text{ W}\end{aligned}$$

Therefore, the power dissipated in 12 Ω resistor is 8.333 W.

Step 7 of 8

Apply current division.

$$\begin{aligned}i_2 &= \frac{10}{10 + 40} i \\&= \frac{10}{10 + 40} (2.5) \\&= 500 \text{ mA}\end{aligned}$$

Therefore, the current i_2 is 500 mA.

By using Ohm's law,

$$\begin{aligned}v_2 &= i_2 R \\&= (500 \text{ mA})(40 \Omega) \\&= 20 \text{ V}\end{aligned}$$

Therefore, the voltage v_2 is 20 V.

Step 8 of 8

Calculate the power dissipated in 40 Ω resistor.

$$\begin{aligned}P_2 &= v_2 i_2 \\&= (20 \text{ V})(500 \text{ mA}) \\&= 10 \text{ W}\end{aligned}$$

Therefore, the power dissipated in 40 Ω resistor is 10 W.